

CS & IT ENGINEERING



Computer Network - 2

Network Layer

Lecture No. - 02

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Recap of Previous Lecture



Topic

Identification Number

Topic

IPv4 Options



Topics to be Covered



Topic

IPv4 Options

Topic

ICMP

Topic

Supernetting

[NAT]

[GATE-2014] [2 Mark]



#Q. Every host in an IPv4 network has a 1-second resolution real-time clock with battery backup. Each host needs to generate up to 1000 unique identifiers per second. Assume that each host has a globally unique IPv4 address. Design a 50-bit globally unique ID for this purpose. After what period (in seconds) will the identifiers generated by a host wrap around ?

Ans = 256

Design a 50-bit globally unique ID.

IPv4 Address : [32 bits]

Identification Number : [18 bits] ✓

Total number of unique identifiers = $[2^{18}]$
 [In each host, before wrap-around]

1000 identifiers → 1 Sec

2^{18} identifiers → $\frac{2^{18}}{1000} \text{ sec}$

→ 262.144 sec

Design a 50-bit globally unique ID.

IPv4 Address : 32 bits

Identification Number : [18 bits]

(1000) identifiers → 1 Sec ⇒ 10 bit counter

Identification Number : 18 bits

Per-second identifier counter : [10 bits] ⇒

Time-based counter : (8 bits) ✓

Wrap-around time for identifiers = $2^8 \text{ sec} = 256 \text{ sec}$



Topic : IPv4 Packet Header



0		16		31	
VER	HLEN	Type of Services	Total Length		
Identification Number			0	D F	M F
Time-to-Live		Protocol Type	Header Checksum		
[Source IPv4 Address (32 - bits)]					
[Destination IPv4 Address (32 - bits)]					
[Optional Header (Options)]					
Payload					



Topic : IPv4 Optional Header





Topic : Option Type

1 byte



Number	Option Type
0	⇒ End of Option [Padding]
1	⇒ No Operation (start/end)
7	✓ Record Route
68	✓ Timestamp
82	Traceroute
→ <u>131</u>	Loose Source Route
<u>137</u>	Strict Source Route



Topic : Source Routing



→ Predetermine a route for the datagram

→ Routing Path is determined by either "Source Host" or "Centralized Server"

→ Routing Path is append in IPv4 options field by Source Host

→ Source routing are of two types :

1. Strict Source Routing
2. Loose Source Routing

Routing Path:
⇒ Source Router
to Dest. Router



Topic : Strict Source Routing

→ Complete path from Source Router to Destination Router
[Sequence of IP address of intermediate routers]

→ Datagram is required to follow exact defined route

→ All of the routers defined must be visited by the datagram

→ if the datagram visits a router that is not on the list
then it discard the datagram and ICMP error message is generated





Topic : Strict Source Routing



(in byte)

Type = 1 00000001	Type = 137 10001001	Total Length = 39 00100111	Pointer = 4 00000100	12
First IP Address (Filled when started) Source Router IP				
Second IP Address (Filled when started) R ₁ (IP)				
Third IP Address (Filled when started) R ₈ (IP)				
⋮				
(g th) Last IP Address (Filled when started) Dest-Router IP				



Topic : Loose Source Routing



→ Similar to the strict source routing, but more relaxed

→ All of the routers defined must be visited by the datagram
[in same order]

→ In addition, the datagram can visit other routers as well



Topic : ICMP

✓ ⇒ Layer-3 protocol



→ Internet Control Message Protocol (ICMP)



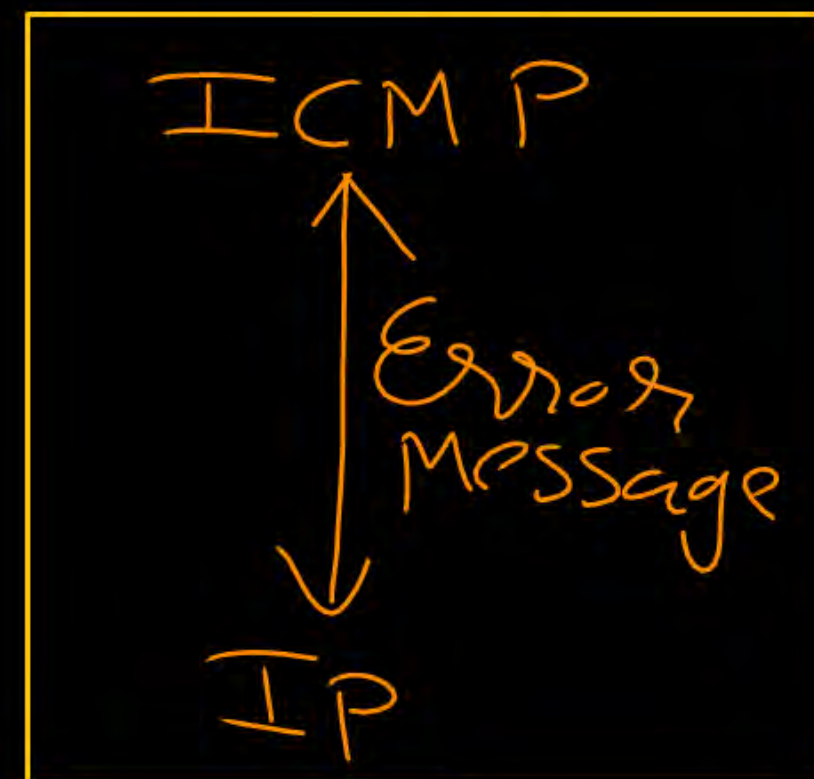
→ To communicate network level information
[Used by Hosts and Routers]

1. Error Reporting ✓

[like Unreachable Network / Host / Protocol / Port]

2. Echo Request / Reply

[Used by "Ping"]



IP, Protocol type = ICMP
IP[ICMP Message]



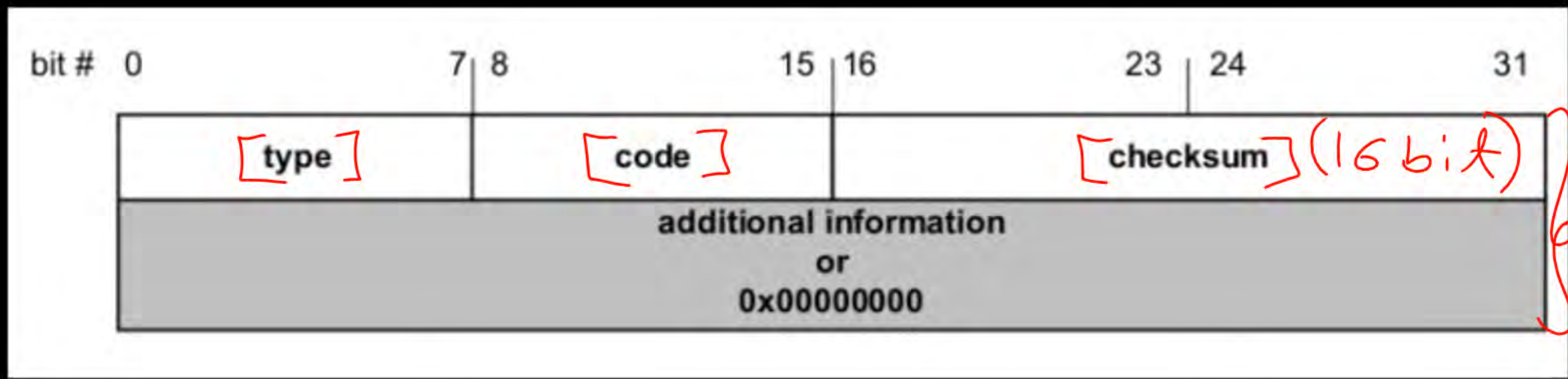
Topic : ICMP



- Network Layer Protocol
- ICMP Messages carried in payload field of IP datagram
- ICMP Message contains :
 1. Type ✓
 2. Code ✓



Topic : ICMP Message Format



Min^m
ICMP
Message
Size
(8 byte)



Topic : ICMP Message Format



- Type (1 byte) : Type of ICMP message
- Code (1 byte) : Subtype of ICMP message
- Checksum (2 byte) : Similar to IP header checksum
[Checksum is calculated over entire ICMP message]
- Each ICMP messages is at least 8 bytes long
[If there is no additional data, 4 bytes set to zero]



Topic : ICMP Message Format



Type	Description
<u>0</u>	Echo Reply
<u>8</u>	Echo Request
<u>3</u>	Destination Unreachable
<u>11</u>	Time Exceeded
<u>30</u>	<u>Traceroute</u>
<u>4</u>	<u>Source Quench</u>

} Ping

} Error Reporting



Topic : Ping



- Application layer utility
- Used to test the reachability of a host on an IP network
- Ping also measures the round-trip time (RTT)
[messages sent from the originating host to a destination that are echoed back to the source]

$$RTT > 2t_p$$

$$RTT \approx 2t_p$$

Host A
[Always client]

Ping Prog.

Host B
IP

Host B
IP

ICMP

ICMP
Echo Req.
Message

ICMP Echo
Reply

IP

IP[ICMP Echo]
Req.

IP[ICMP
Reply]

Host B
client/server/
Networking Device



Ping Prog.

Host A
IP

Host A
IP

ICMP

ICMP Echo
Reply MSG

ICMP Echo
Req. Message

IP

IP[ICMP
Reply]

IP[ICMP Echo]
Req

Example :-

[MSQ]



#Q. Identify correct statement(s) regarding ping utility.

- ✓ ☒ A Ping measure round-trip time.
- ✓ ☒ B Ping report packet loss.
- ✓ ☒ C Ping is used to test network connectivity.
- ☐ D Ping is a client-server application. CS-10

Ans: A, B & C



Topic : ICMP Error Message

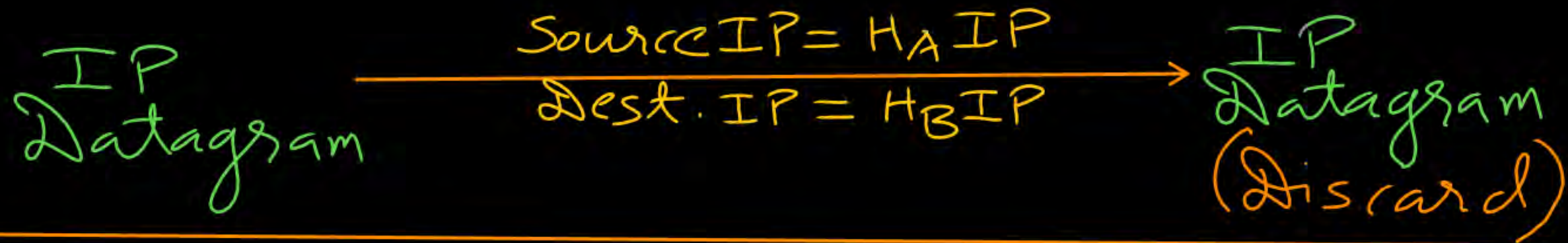


→ ICMP error messages report error conditions

→ Whenever IP datagram is discarded at Destination Host or Intermediate Router it sent "ICMP Error Message" to Source Host

→ ICMP error messages contains complete IP header and the first 8 bytes of the IP datagram payload

Source Host ... $\bar{R}_1$ $\bar{R}_2$ $\bar{R}_3$... Router
Host A R R R R



ICMP
 $\uparrow$
ICMP Error
 $\uparrow$
IP

IP [ICMP Error]
Source IP = R(IP)
Dest IP = $H_A \text{ IP}$

ICMP
 $\downarrow$
ICMP Error Message
 $\downarrow$
IP



Topic : Destination Unreachable

Type = 3

Code	Description
0	Network Unreachable ✓
1	Host Unreachable ⇒ Dest. Router
2	Protocol Unreachable ⇒ Dest. Host (IP)
3	Port Unreachable ⇒ Dest. Process does not exist
4	Fragmentation Required DF Flag "Set"



Topic : Traceroute



→ Command-line utility

→ Used to trace the route of network packets
[from source host to destination host]

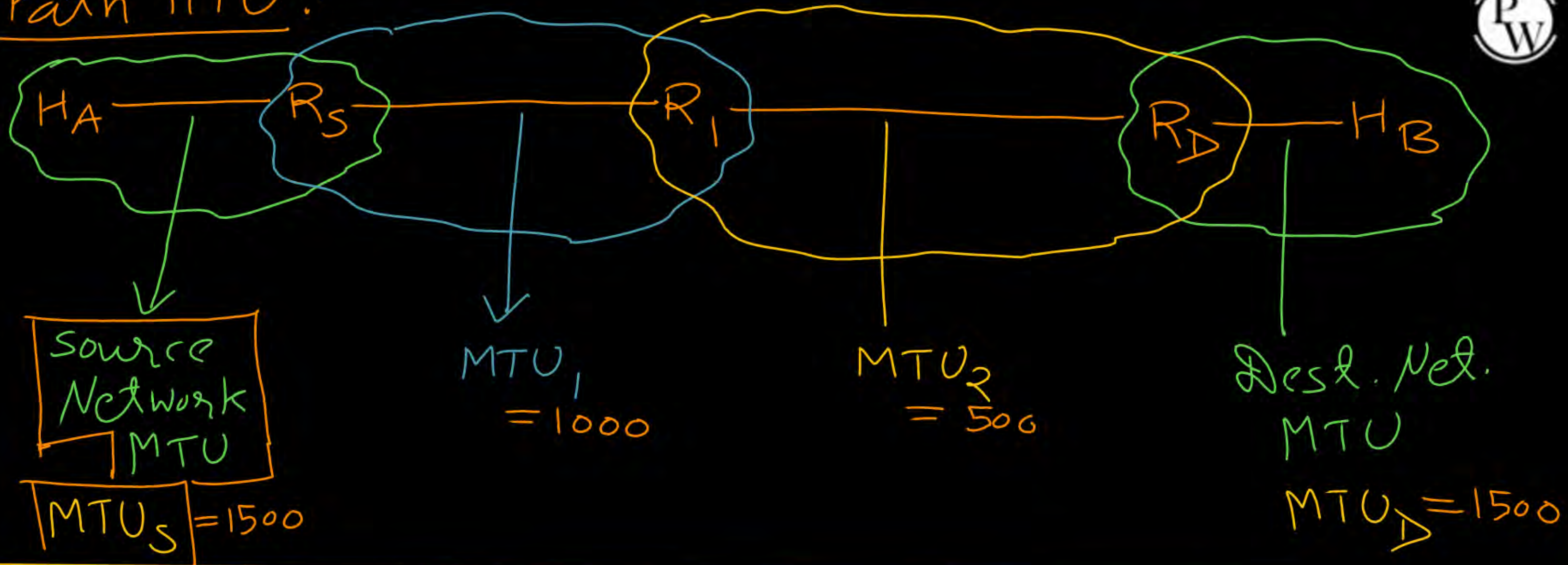
→ Identifies the IP addresses of routers along the path

→ Valuable tool for troubleshooting network connectivity issues

#Q. Trace-route reports a possible route that is taken by packets moving from some host A to some other host B. which of the following options represents the technique used by traceroute to identify these hosts?

- A** By progressively querying routers about the next router on the path to B using ICMP packets, starting with the first router.
- B** By requiring each router to append the address to the ICMP packet as it is forwarded to B. The list of all routers en-route to B is returned by B in an ICMP reply packet.
- C** By ensuring that an ICMP reply packet is returned to A by each router en-route to B, in the ascending order of their hop distance from A
- D** By locally computing the shortest path from A to B

Path MTU :-



$$\text{Path MTU} = \text{Min}[MTU_S, MTU_1, MTU_2, MTU_D]$$



2 mins Summary



Topic

IPv4 Options ✓

Topic

ICMP ✓

Topic

~~Supernetting~~

CN-2 (Important)
⇒ Application layer
Protocol
⇒ CSMA/CD
 $T_x \geq 2t_p$



THANK - YOU

